

5G/6G COMMUNICATION-ECA5604

EXPT-7

Analog Beamforming Analysis Using MATLAB

Aim:

To analyze the performance of analog beamforming by studying beamforming gain, beam width,

and main lobe direction using MATLAB.

Objectives

1. To analyze the beamforming gain achieved by varying the number of antenna elements in an analog beamforming system using MATLAB.
2. To compare the beam width of an antenna array for different beamforming configurations using MATLAB.
3. To analyze the main lobe direction by steering the antenna array beam to different angles using MATLAB.

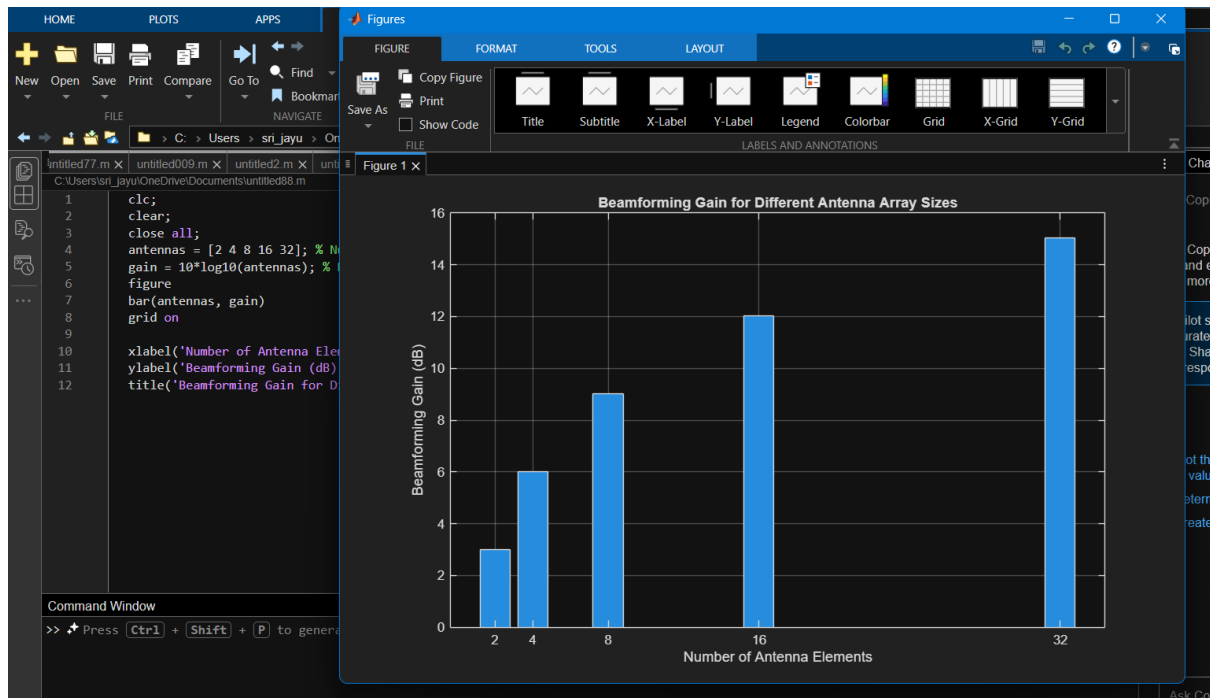
Software Required:

- MATLAB

MATLAB Code:

```
clc;  
clear;  
close all;  
antennas = [2 4 8 16 32]; % Number of Antenna Elements  
gain = 10*log10(antennas); % Beamforming Gain (dB)  
figure  
bar(antennas, gain)  
grid on  
  
xlabel('Number of Antenna Elements')  
ylabel('Beamforming Gain (dB)')
```

title('Beamforming Gain for Different Antenna Array Sizes')



Objective – 2

Compare Beam Width (Degrees)

Analyze the beam width corresponding to different antenna array sizes using MATLAB.
Observe

how increasing the number of antenna elements results in a narrower beam, thereby improving spatial resolution.

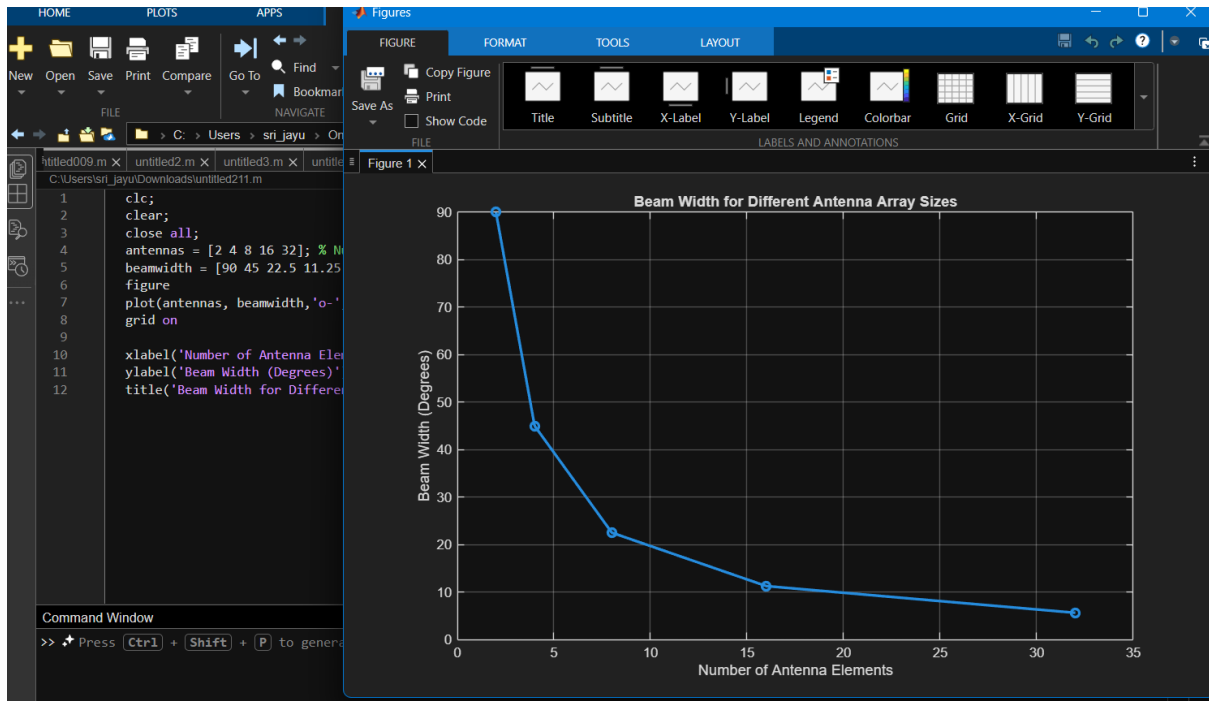
MATLAB Code:

```
clc;  
clear;  
close all;  
antennas = [2 4 8 16 32]; % Number of Antenna Elements  
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)  
figure  
plot(antennas, beamwidth,'o-','LineWidth',2)  
grid on
```

```

xlabel('Number of Antenna Elements')
ylabel('Beam Width (Degrees)')
title('Beam Width for Different Antenna Array Sizes')

```



Objective – 3

Analyze Main Lobe Direction (Degrees)

Simulate beam steering by varying the steering angle of the antenna array using MATLAB.

Observe the change in the main lobe direction and understand how analog beamforming directs

energy toward the intended user.

MATLAB Code:

```

clc;

clear;

close all;

theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)

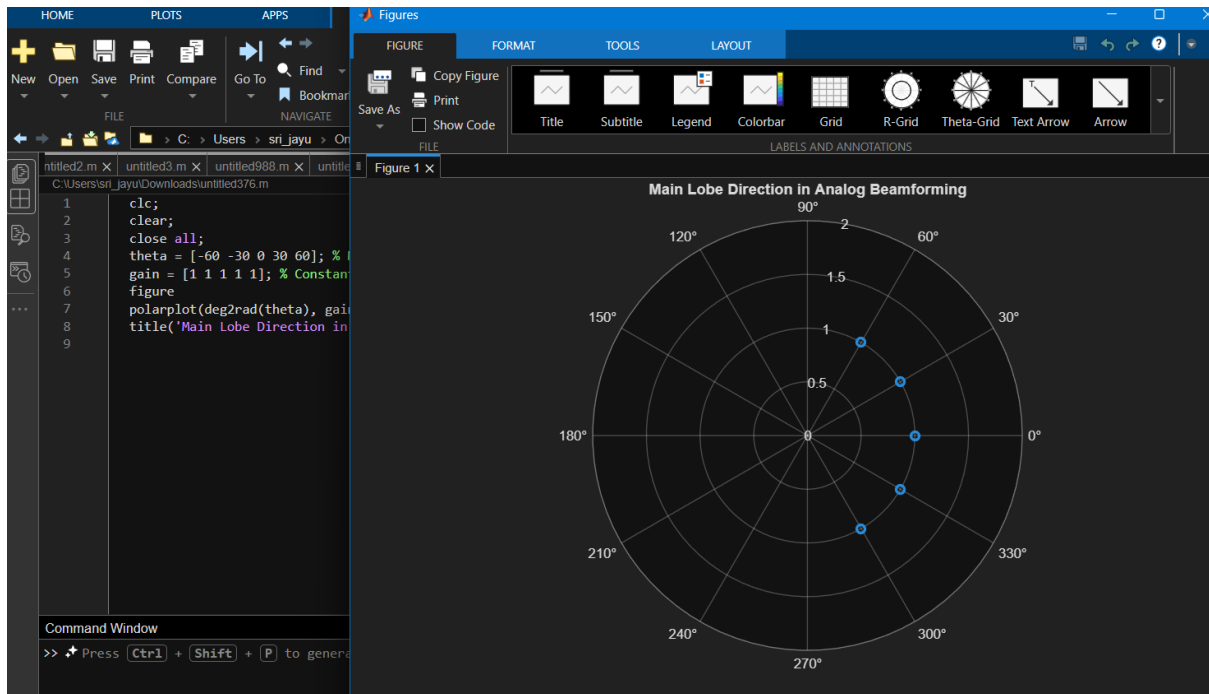
gain = [1 1 1 1 1]; % Constant normalized gain

figure

```

```
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)

title('Main Lobe Direction in Analog Beamforming')
```



Result

The MATLAB analysis successfully demonstrated the characteristics of analog beamforming. The

results showed that increasing the number of antenna elements improves beamforming gain and

reduces beam width, while phase adjustment enables the main lobe to be steered toward different

directions.